Google Cloud

Trust and Security with Google Cloud

This study guide covers the materials from the fifth course of the six-course series for the Cloud Digital Leader certification.

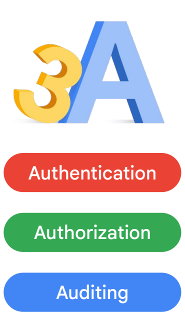
Certification exam focus

The Cloud Digital Leader certification exam is designed to assess knowledge of cloud literacy, common business use cases, and how cloud solutions and the capabilities of Google Cloud products support an enterprise.

Cyber attacks and threats

- **Social engineering:** Using deceptive emails (phishing) to trick people into downloading malware, sharing passwords, or exposing data.
- **Malware, viruses, and ransomware:** Malicious software that aims to disrupt operations, inflict damage, or gain unauthorized access to computer systems.
- **DDoS attacks:** Flooding a service with illegitimate traffic from compromised systems, denying access to genuine users.
- **Large language model (LLM) attacks:** Targeting generative AI (gen AI) to manipulate output, extract sensitive data, bypass safety guardrails, or abuse computing resources.

The three As of cloud identity management



- **Authentication:** Presenting unique credentials, such as passwords, physical tokens, or biometric data like fingerprints or voice recognition.
- **Authorization:** Determining what a user or system is allowed to do within a system.
- **Auditing/accounting:** Monitoring and tracking user activities within a system.

Concepts related to securing sensitive data

- **Privileged access:** Grants specific users access to a broader set of resources than ordinary users.
- **Least privilege:** Advocates granting users only the access they need to perform their job responsibilities.
- **Zero-trust architecture:** Assumes that no user or device can be trusted by default.

Concepts related to mitigating cyber threats

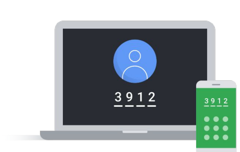
- **Security by default:** Emphasizes integrating security measures into systems and apps from the initial stages of development.
- **Security posture:** The overall security status of a cloud environment, measured against defined baselines.
- **Cyber resilience:** An organization’s ability to withstand and recover quickly from cyber attacks.

Concepts related to securing cloud resources

- **Firewall:** A network device that regulates traffic based on predefined security rules.
- **Data Loss Prevention:** Strategies and tools to automatically identify, monitor, and protect sensitive information.

Two-step verification (2SV)

A security feature that requires users to provide two different forms of identification to log in.



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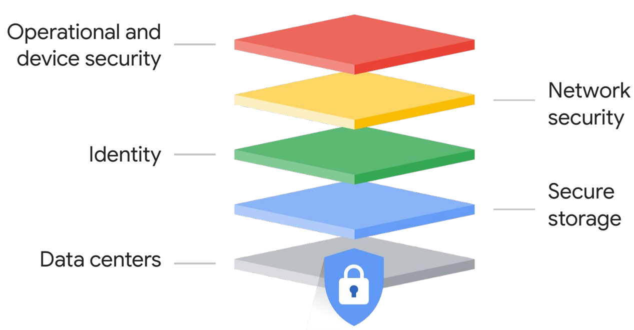
Cloud security components

- **Control:** The policies and technical safeguards used to mitigate security risks.
- **Compliance:** The adherence to laws, regulations, and industry standards.
- **Confidentiality:** Keeping important information safe and secret.
- **Integrity:** Keeping data accurate and trustworthy.
- **Availability:** Ensuring cloud systems and services are accessible and ready for use by the right people when needed.

Google's multilayered security approach



Encryption and decryption

- **Encryption:** The process of converting data into an unreadable format by using an algorithm and a secret key.

  Data encryption in different states: In use, in transit, at rest.

- **Decryption:** The use of an encryption key to restore encrypted data back to its original form.

Google Threat Intelligence

A Google solution that provides curated data on malicious actors, indicators of compromise, and known attack patterns. It's powered by five unique data sources:

Google Threat Intelligence

1. Frontline intelligence
2. Crowdsourced threat intelligence
3. Open-source threat intelligence (OSINT)
4. Human-curated threat intelligence
5. Google insights

Security Command Center

A centralized security management and risk platform designed to protect cloud workloads. It provides:

- **Posture management:** It discovers cloud assets, scans for misconfigurations, and continuously monitors compliance against major industry standards.
- **Vulnerability detection:** It identifies weaknesses in virtual machines, containers, and web apps before they can be exploited.
- **Threat detection:** It monitors system logs and runtime environments in real time to detect active threats like malware, unauthorized access, and resource abuse.

Security operations (SecOps)

The continuous collaboration between security and IT teams to monitor, detect, and mitigate cyber threats in real time.

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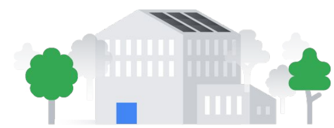
Google security offerings

- Cloud VPN
- Cloud Interconnect
- Cloud Armor
- Cloud Logging
- Identity and Access Management (IAM)
- Confidential Computing
- Certificate Manager
- Identity-Aware Proxy

Google-built global network and data centers

- **Physical security:** Access is managed through multiple physical layers, including custom-designed electronic access cards, biometric scanners, laser barriers, and 24/7 human security monitoring.
- **Networking isolation**: The global network is a purpose-built private fiber network that handles all customer traffic. This dramatically reduces exposure to the public internet and enables fast, secure communication across the globe.



Google's AI security offerings

- **Gemini in Google SecOps:** An advanced gen AI-powered feature integrated into the Google SecOps platform that acts as a virtual security analyst, dramatically speeding up threat detection, investigation, and response workflows.
- **AI Protection**: Integrated with Security Command Center, it's a set of capabilities designed to safeguard AI workloads and data across clouds and models, irrespective of the platforms you choose to use.
- **Model Armor:** A fully managed Google Cloud security service designed to enhance the security and safety of LLM apps, particularly gen AI models.

Google Cloud trust principles

1. You own your data, not Google.
2. Google does not sell customer data to third parties.
3. Google Cloud does not use customer data for advertising.
4. All customer data is encrypted by default.
5. We guard against insider access to your data.
6. We never give any government entity "backdoor" access.
7. Our privacy practices are audited against international standards.

Google provides valuable insights and accountability through transparency reports and third-party audits, which shed light on government and corporate actions that affect privacy, security, and access to information.

Concepts about storing and keeping data secure

- **Data sovereignty:** The legal concept that data is subject to the laws and regulations of the country where it resides.
- **Data residency:** The physical location where data is stored or processed.